

FleetOps

FleetOps operates many Ignition gateways as one fleet: an agent on each gateway reports its real state, a console holds desired state, and the difference is drift.

Version 0.2.0 Ignition 8.1.19+ and 8.3 Runs on: Host (console) and every fleet gateway (agent)

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1. Overview

FleetOps operates many Ignition gateways as one fleet. An agent module on each gateway reports its real state (module versions, project hash, certificates, backups); a console module at the center holds what each gateway is supposed to be, as a signed desired-state document, and reports the difference as drift. A signed channel carries desired state and a closed, allowlisted set of imperative actions from console to agents.

FleetOps is not a general remote-execution tool: only allowlisted actions run, and each requires a valid signature.

2. What You Need

- A SparkBridge Host gateway to run the console role.
- The agent role installed on every gateway that is a fleet member.
- The signed FleetOps module from the SparkBridge bundle.
- A P-256 signing key pair for issuing desired state and signed actions.

3. Install

FleetOps carries its own license. Install the signed module through Config > Modules > Install or Upgrade a Module; no build step is required. Install the agent role on every fleet member and the console role on the gateway that will act as the center.

4. License

See the README in the bundle for licensing detail.

5. Configure

A JVM system property wins over the properties file; otherwise settings come from `<dataDir>/fleetops/fleetops.properties`. Secrets are the exception: the broker password and REST token come only from the file named by `fleetops.secrets.file`, which must be mode 600.

<code>fleetops.agent.enabled</code>	false by default; a module dropped on a gateway must not silently join a fleet.
<code>fleetops.console.enabled</code>	false by default.
<code>fleetops.broker.uri</code>	default <code>tcp://127.0.0.1:1883</code> .
<code>fleetops.trustedkey</code>	base64 P-256 public key. Absent means the agent reports and accepts nothing.
<code>fleetops.action.allowlist</code>	comma-separated action names; empty means no action is permitted.
<code>fleetops.action.certReplaceEnabled</code>	false by default; second explicit local gate for the private-key-bearing CERT_REPLACE action.
<code>fleetops.agent.intervalSeconds</code>	default 60, floor of 5; this is also the drift-detection latency.

Full property list is in the module README's Configuration section. Desired-state templates let the same shape be applied to many sites at once via `apply-template`, instead of hand-writing one properties document per gateway.

6. Operate

The agent reports on `fleetops.agent.intervalSeconds`, which is also the drift-detection latency (floor 5 seconds, default 60). The console evicts fleet rows that go stale relative to when the agent actually collected them (TTL on `collectedAt`, not on receipt); use the `evict` CLI command to retract retained state for a renamed or decommissioned site by hand. Watch the drift report for module-version, configuration, certificate-expiry and backup-status differences from desired state.

7. Verify

Use the operator CLI to publish desired state and issue a signed action, then confirm the fleet converges:

```
java -jar ... publish-desired --key fleet-signing.key --site Austin/Site12 \
  --file desired.properties --version 2026-08-16.1 --broker tcp://host:1883
java -jar ... watch --broker tcp://host:1883 --seconds 30 --report drift
```

Full multi-site rig proof is in `docs/LIVEPROOF.md` and `docs/rig-transcript.txt` in the repo.

8. Troubleshoot

Symptom	Cause	Fix
Agent reports nothing to the console	<code>fleetops.trustedkey</code> not set on the agent	Set the console's public key on the agent; absent, the agent accepts nothing
Signed action rejected	Action name not in <code>fleetops.action.allowlist</code>	Add the action name to the allowlist on the target gateway

Symptom	Cause	Fix
Certificate replacement refuses to run	<code>fleetops.action.certReplaceEnabled</code> is false (default)	Set it true only after accepting that the action carries a private key
Site shows as stale in the console	Agent has not reported within the TTL window	Check the agent's <code>fleetops.agent.intervalSeconds</code> and network path to the broker; use <code>evict</code> to retract a decommissioned site

9. Reference

- Agent poll floor: 5 seconds; default 60 seconds.
- Secrets file must be mode 600.
- Download: [FleetOps 0.2.0, signed module](#).

10. Support

[FleetOps in the docs](#) · [FleetOps on the Host product page](#) · [Contact](#) · [GitHub releases](#)